AWS Cloud Practitioner Exam Study Guide Summary

This summary condenses the key information from your AWS Cloud Practitioner Study Guide, focusing on essential concepts, services, and exam-relevant details to help you prepare effectively.

Domain 1: Cloud Concepts (24%)

1.1 Benefits of AWS Cloud

- **Cloud Computing Criteria**: On-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service.
- **Benefits of Cloud Computing**: Flexibility, security, scalability, cost savings (trading fixed for variable expense), increased speed and agility, stopping guessing about capacity, and global reach in minutes.
- **AWS Global Infrastructure**: A collection of interconnected infrastructures (Regions, Availability Zones, Edge Locations) designed for resilience and high availability.
  - **High Availability**: Designing for minimal downtime and quick, automatic recovery.
  - **Fault Tolerance**: Designing for zero downtime, with active-active redundant systems.
  - **Disaster Recovery**: Planning to operate through a disaster.
  - **Scaling**: Ability to increase or decrease system load.
    - **Vertical Scaling**: Resizing an instance to a larger size (more CPU/memory).
    - **Horizontal Scaling**: Adding more instances of the same size to handle load, often with a load balancer.
  - **Elasticity**: Automation with horizontal scaling to match capacity with demand.

1.2 Design Principles of the AWS Cloud (AWS Well-Architected Framework)

- **Pillars**:
  - **Operational Excellence**: Run workloads effectively, gain insight into operations, and improve processes.
    - Principles: Perform operations as code, make frequent small reversible changes, refine procedures, anticipate failures, learn from failures.
  - **Security**: Protect data, systems, and assets.
    - Principles: Implement strong identity foundation, enable traceability, apply security at all layers, protect data in transit and at rest, keep people away from data, prepare for security events.
  - **Reliability**: Perform workload function correctly and consistently.
    - Principles: Automatically recover from failure, test recovery procedures, scale horizontally, stop guessing capacity, manage change in automation.

    - **Performance Efficiency**: Use computing resources efficiently.
        - Principles: Democratize advanced technologies, go global in minutes, use serverless architectures, experiment more often, consider mechanical sympathy.
    - **Cost Optimization**: Run systems to deliver business value at the lowest price.
    - **Sustainability**: Focus on environmental impact, energy consumption, and efficiency.
        - Principles: Understand impact, establish sustainability goals, maximize utilization, adopt efficient hardware/software, use managed services, reduce downstream impact.

1.3 Benefits of and Strategies for Migration to the AWS Cloud

- **AWS Cloud Adoption Framework (CAF)**: Provides a comprehensive approach for cloud transformation, identifying organizational capabilities for success. Benefits include reduced business risk, increased performance, and enhanced security
- **Seven Migration Strategies (The 7 Rs)**:
    1. **Retire**: Decommission or retire applications no longer needed.
    2. **Retain**: Keep applications in the source environment if not ready to migrate.
    3. **Rehost (Lift and Shift)**: Migrate applications without changes.
    4. **Relocate**: Large-scale server migration using AWS services like AWS Outposts.
    5. **Repurchase (Drop and Shop)**: Replace existing applications with a different version or product, often SaaS.
    6. **Replatform (Lift, Tinker, and Shift)**: Migrate with some optimization for cloud efficiency.
    7. **Refactor/Re-architect**: Re-imagine how an application is architected to take full advantage of cloud-native features for agility, performance, and scalability.

1.4 Concepts of Cloud Economics

- **Total Cost of Ownership (TCO)**: Includes operational expenses (day-to-day), capital expenses (one-time purchases), labor costs, and software licensing costs.

Domain 2: Security and Compliance (30%)

2.1 AWS Shared Responsibility Model

- **AWS is responsible for security *OF* the cloud**: This includes the global infrastructure (hardware, software, networking, facilities).
- **Customer is responsible for security *IN* the cloud**: This includes customer data, platform applications, identity and access management, operating systems, network and firewall configurations, and client-side data encryption.

2.2 AWS Cloud Security, Governance, and Compliance Concepts

- **Compliance Information**: Varies by service and location/industry. Found in AWS Artifact.
- **Security Services**:

    - **AWS WAF (Web Application Firewall)**: Protects applications from common exploits (SQL injection, cross-site scripting).
    - **Amazon GuardDuty**: Threat detection service monitoring for malicious activity and unauthorized behavior.
    - **AWS Shield**: Managed DDoS protection service.
    - **Encryption**:
        - **At Rest**: Stored data encryption.
        - **In Transit**: Data encryption as it moves between locations.
- **Monitoring & Auditing Services**:
    - **Amazon CloudWatch**: Monitoring and collecting operational data.
    - **AWS CloudTrail**: Logs events related to AWS resource creation and management; provides governance, compliance, operational auditing, and risk auditing.
    - **AWS Config**: Takes inventory of configurations and audits resources to ensure correct configurations.

2.3 AWS Access Management Capabilities

- **Principle of Least Privilege**: Granting users only the minimum access needed to perform their work.
- **AWS Account Root User**:
    - The first identity created with a new AWS account.
    - Has complete and unrestricted access.
    - **Should NOT be used for daily tasks**.
    - Tasks requiring root user: change account settings, restore IAM user permissions, activate IAM access to Billing, view tax invoices, close AWS account, register as seller, configure S3 with MFA, edit/delete bucket policies, sign up/request access key for AWS GovCloud.
- **AWS IAM (Identity and Access Management)**: Securely manage access to AWS resources.
    - **IAM Users**: Associated with usernames/passwords, access keys for programmatic access, MFA enabled, password complexity/rotation enforced.
    - **IAM Groups**: Collections of IAM users; users inherit group permissions.
    - **IAM Roles**: Temporary credentials assumed by entities (users, programs, services) to gain temporary permissions; used for cross-account access.
    - **IAM Policies**: Documents that grant or deny permissions. Can be managed (AWS-managed or customer-managed) or unmanaged (inline)
        - **IAM Policy Simulator**: Used to test and troubleshoot IAM and resource-based policies.
        - **Bucket Policies**: Resource-based policies that differ from IAM roles. MFA Delete adds protection to S3 buckets.

2.4 Components and Resources for Security

- **Network Access Control Lists (NACLs)**:
    - Operate at the subnet level for traffic entering or leaving a subnet.

    - **Stateless**: Inbound and outbound rules are separate (if inbound allowed, outbound must also be allowed explicitly).
    - Can explicitly deny traffic.
- **Security Groups**:
    - Operate at the resource level (e.g., EC2 instances, RDS instances).
    - **Stateful**: If inbound traffic is allowed, outbound traffic is automatically allowed.
    - **Cannot explicitly deny** traffic; for explicit denial, use NACLs.
    - Default behavior: Denies all inbound traffic, allows all outbound traffic.
- **AWS Trusted Advisor**: Provides recommendations around security, cost optimization, performance, service limits, and fault tolerance.
- **Amazon Inspector**: Automated security assessment service.
- **AWS Marketplace**: Allows finding and deploying third-party solutions.

Domain 3: Cloud Technology and Services (34%)

3.1 Methods of Deploying and Operating in the AWS Cloud

- **Deployment Methods**: Programmatic access (APIs), AWS Command Line Interface (CLI), AWS Management Console (GUI), Infrastructure as Code (IaC).
- **Cloud Deployment Models**:
    - **Cloud Native**: Applications built specifically for the cloud.
    - **Hybrid Cloud**: Combines public cloud and on-premises infrastructure.
    - **On-premises**: Infrastructure hosted locally.
    - **Public Cloud**: Services available to the public over the internet.
    - **Private Cloud**: Dedicated cloud services for a single organization, often on-premises (e.g., AWS Outposts).
    - **Multi-Cloud**: Using more than one public cloud provider (e.g., AWS and Azure).

3.2 AWS Global Infrastructure

- **AWS Region**: A geographic area with two or more Availability Zones. Data resides in the region unless explicitly moved.
    - **Region Selection Criteria**: Compliance, available services, pricing, proximity to customers.
    - **Exam Tip**: Choose the closest region for lower latency and check compliance.
- **Availability Zone (AZ)**: One or more discrete data centers designed to be isolated from failures in other AZs within the same region.
    - Services tied to a specific AZ (e.g., Amazon EBS, EC2 instances) are AZ-resilient, meaning if that AZ fails, the service might fail.
- **Edge Locations (Points of Presence)**: Sites used to cache copies of content to reduce latency for users.
    - Used by services like **Amazon CloudFront** (CDN for static and dynamic content caching) and **AWS Global Accelerator** (improves performance over TCP/UDP for applications, provides static IPs, no caching). Both integrate with AWS Shield for DDoS protection.
- **Cloud Models**:

    - **IaaS (Infrastructure as a Service)**: Provides virtualized computing resources (e.g., EC2).
    - **PaaS (Platform as a Service)**: Provides a platform allowing customers to develop, run, and manage applications without the complexity of building and maintaining infrastructure.
    - **SaaS (Software as a Service)**: Software applications hosted by a third-party and made available to customers over the Internet.
    - **DaaS (Database as a Service)**: Managed database services.

3.3 AWS Compute Services

- **Amazon EC2 (Elastic Compute Cloud)**: Virtualization as a service (IaaS), providing resizable compute capacity. Instances are AZ-resilient.
    - **Instance Store**: Temporary, local storage. Data is lost if the instance moves or stops.
    - **EC2 Instance Types**:
        - **General Purpose**: Balance of compute, memory, and networking.
        - **Compute Optimized**: High-performance processors, good for compute-intensive workloads.
        - **Memory Optimized**: Fast performance for workloads processing large datasets in memory.
        - **Accelerated Computing (F1)**: Hardware accelerators (GPUs, FPGAs) for specialized tasks.
        - **Storage Optimized (D2)**: For high-volume data workloads.
        - **Burstable Instances**: Low CPU loads that can burst to higher levels with burst credits, cost-effective.
    - **Amazon Machine Image (AMI)**: Provides the information required to launch an instance. Can create custom AMIs.
- **Container Services**:
    - **Amazon ECS (Elastic Container Service)**: Managed container orchestration service for Docker containers.
    - **Amazon EKS (Elastic Kubernetes Service)**: Allows running Kubernetes on AWS.
- **AWS Lambda**: Serverless compute service (Function as a Service - FaaS). Runs code without provisioning or managing servers; event-driven.
- **High Availability & Scalability**: Achieved using **Auto Scaling Groups (ASG)** to scale EC2 instances automatically based on criteria and **Load Balancers (ELB)** to distribute incoming connections across multiple targets

3.4 AWS Database Services

- **Amazon Relational Database Service (RDS)**: Managed relational database service (DBaaS). Supports MySQL, MariaDB, PostgreSQL, Oracle, Microsoft SQL Server, and Amazon Aurora.
    - Designed for **OLTP (Online Transaction Processing)**.

        - **Multi-AZ deployments**: For high availability, automatically creates a standby replica in a different AZ.
- **Amazon Aurora**: MySQL and PostgreSQL compatible relational database with high performance and availability.
    - **Aurora Serverless**: Doesn't require provisioning or managing instances; scales capacity automatically based on demand using Aurora Capacity Units (ACUs).
    - **Aurora Global Database**: For global resilience.
- **Amazon DynamoDB**: Fully managed NoSQL database service. Highly resilient across multiple AZs and supports global tables. [100]
    - Supports provisioned capacity or on-demand mode for scaling. [101]
    - **DAX (DynamoDB Accelerator)**: In-memory cache for DynamoDB, designed for latency-sensitive and high-read workloads. [102]
- **AWS Snow Family**: For large-scale data transfer.
    - **SnowCone**: 8 TB data transfer. [103]
    - **Snowball Edge**: Data migration and edge computing device (supports EC2 instance types and Lambda functions). [104]
    - **Snowmobile**: For petabytes to exabytes of data. [105]
- **AWS Database Migration Service (DMS)**: Capable of data migration and schema conversion between different database engines. [106106106106]
- **AWS Schema Conversion Tool (SCT)**: Helps transform database schemas for migration. [107]
- **Amazon Elastic Block Store (EBS)**: Provides persistent block storage volumes for EC2 instances. AZ-resilient. [108108108108]
- **Amazon Elastic File System (EFS)**: Scalable, elastic file storage for use with AWS Cloud services and on-premises resources. [109]

3.5 Networking and Content Delivery

- **Amazon VPC (Virtual Private Cloud)**: A logically isolated section of the AWS Cloud where you can launch AWS resources in a virtual network that you define. [110110110]
    - **Default VPC**: One per region, configured by AWS. [111]
    - **Custom VPC**: Configured by the user, isolated and private by default. [112]

- **Subnets**: Logical subdivisions of a VPC's IP address range, used for security, isolation, and development. Can be public (with route to Internet Gateway) or private.[113113113113113113113113113]
- **Internet Gateway (IGW)**: Regional, highly available service allowing communication between your VPC and the internet.[114114114114]
- **NAT Gateway (Network Address Translation Gateway)**: Allows private subnet resources to access the internet or other AWS public services without being directly accessible from the internet.[115]
- **VPC Peering**: Connects two VPCs to allow direct communication using private IP addresses. Can span AWS accounts and regions.[116]
- **VPC Endpoints**: Allows instances in a VPC to connect to AWS public services without requiring an Internet Gateway or NAT Gateway.[117]
  - **Gateway Endpoints**: For S3 and DynamoDB.[118]
  - **Interface Endpoints**: For all other services (uses DNS).[119]
- **AWS PrivateLink**: Provides private connectivity between VPCs, AWS services, and on-premises applications without exposing data to the public internet.[120]
- **AWS Direct Connect**: A dedicated, private network connection from your on-premises data center to AWS. Offers improved performance and enhanced security compared to internet-based connections.
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- **AWS VPN**: Creates an encrypted connection over the public internet.[122]
- **Amazon Route 53**: Highly available and scalable cloud DNS (Domain Name System) web service. Global service, tolerates regional failures.[123123123123]

Domain 4: Billing, Pricing, and Support (12%)

4.1 Concepts of AWS Cloud Financial Management

- **Pricing Models**:
  - **Pay-as-you-go**: Pay only for the services you consume.[124]
  - **On-Demand**: No upfront costs, pay for compute capacity by the hour or second.
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  - **Reserved Instances (RIs)**: Significant discount (up to 75%) compared to On-Demand pricing in exchange for committing to a one-year or three-year term.

    - **Savings Plans**: Flexible pricing model offering lower prices on Amazon EC2, AWS Fargate, and AWS Lambda usage in exchange for a commitment to a consistent amount of usage (measured in $/hour) for a 1-year or 3-year term.
- **Cost Optimization**: Leveraging pricing models, rightsizing resources, and using managed services.
- **AWS Budgets**: Allows you to set custom budgets to track your cost and usage from a simple dashboard.
- **AWS Cost Explorer**: A free tool that allows you to view and analyze your AWS costs and usage.

4.2 AWS Pricing Policies and Billing Practices

- **Factors influencing cost**: Compute, storage, data transfer out. Data transfer *in* is generally free.
- **AWS Organizations**: Allows you to consolidate multiple AWS accounts into an organization that you create and centrally manage.
    - **Consolidated Billing**: All accounts in an organization are paid by one master account, often resulting in volume discounts.

4.3 AWS Support Resources

- **AWS Support Plans**: Different levels of support (Basic, Developer, Business, Enterprise) offering varying response times, technical support, and resources.
- **AWS Documentation**: Comprehensive information on all AWS services, including best practices and whitepapers. [126]
- **AWS Knowledge Center**: A resource for finding answers to questions. [127]
- **AWS Security Center/Blog/Forum**: Resources for security-related information. [128]

Review Questions & Answers:

Here are some review questions and answers from your study guide that are highly relevant for the exam:

- **Which migration strategy moves the existing infrastructure to the cloud as is?**
    - a. Rehost [129]
- **Which IAM identities can have attached permission policies? (Select TWO.)**
    - a. Groups and Users [130]
- **AWS offers pay-as-you-go or On-Demand pricing for many of its services. This variable cost is spread across thousands of customers, resulting in a lower cost than you can get on your own. Which advantage of AWS is this an example of?**
    - a. Benefit from massive economies of scale [131]

- **What is the name of the first identity created with each new AWS account?**
  - a. AWS root user [132]
- **A company wants to build a database where they can automatically scale to meet spikes in read requests. They don't want to have to manage read replicas. Which AWS service is best fit to meet this need?**
  - a. Amazon DynamoDB [133]
- **What describes an AWS edge location?**
  - a. A site that is used to cache copies of your content [134]
- **Which fully-managed service can be used to create dashboards and visualizations based on business intelligence data?**
  - a. Amazon QuickSight (not explicitly in the provided text, but it's the correct answer for this type of question related to BI dashboards. The document mentions "Amazon QuickSight" in a review question snippet that was not expanded in the fetched content, but this is a common service to know for the exam. I will keep the original answer as provided in the PDF output.)
- **A company needs a high-speed dedicated connection to the AWS Cloud. Which service is recommended?**
  - a. AWS Direct Connect [135]
- **An application pulls 490 GB of data from industrial machines nightly. This application processes, filters, and aggregates the data. The result is sent to an Amazon Relational Database Service (Amazon RDS) database. Which Amazon EC2 instance type would BEST meet the requirements of this application?**
  - a. Compute optimized [136]
- **How is a public subnet created?**
  - a. Create a subnet. Create a route in the subnet's route table with the target of the internet gateway. [137]